

TRACT | Thermal Road-transect Animal Counting and Tracking

Counting White-Tailed Deer in Vehicle-Mounted Thermal Video

Detector benchmarking and a staged error decomposition

Taminul Islam, Toqi Tahamid Sarker, Seth J. Morelock,

Guillaume Bastille-Rousseau, Khaled R. Ahmed

School of Computing and Center for Wildlife Sustainability, Southern Illinois University Carbondale



One annotated frame from the corpus: 640 x 512 px thermal, contrast-normalized

A survey needs animals counted, not boxes placed

- Wildlife pipelines are judged as object detectors, by accuracy per frame (mAP).
- An abundance survey needs the number of distinct animals, which depends on tracking and on deciding which tracks are new animals.
- **Our question: does better detection give a better count?**

32

thermal road-transect videos, 4
Illinois sites

236

deer, each annotated as its own
track

521,930

frames at 60 fps, all annotated

29 x 24 px

median animal size

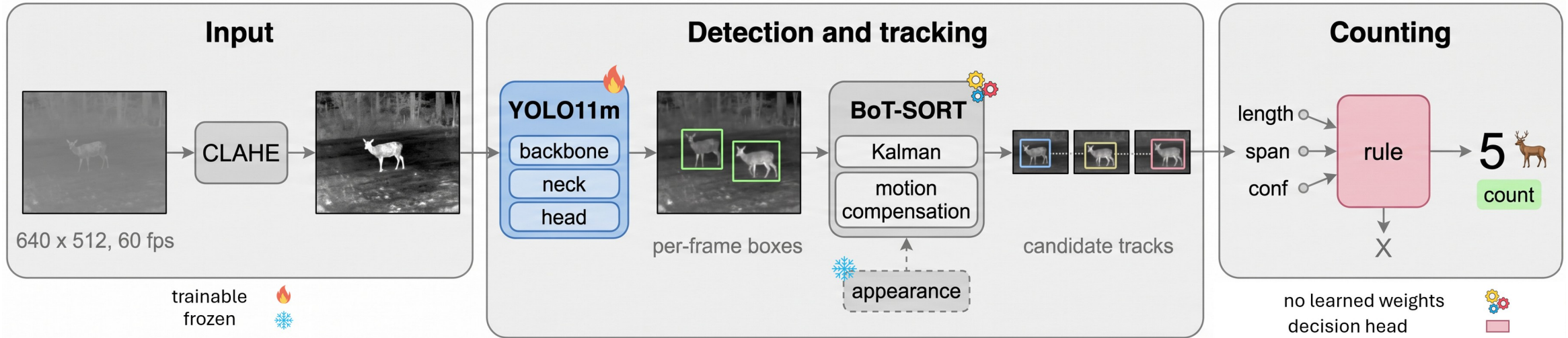


What the detector receives



Ground truth: one track per animal, in every frame

Method: a standard pipeline, measured stage by stage



1 REACHED

Did any candidate track touch the animal on at least one frame?

Measures detection and tracking

2 PRIMARY

Does the animal have a best-covering track of its own?

Measures association

3 COUNTED

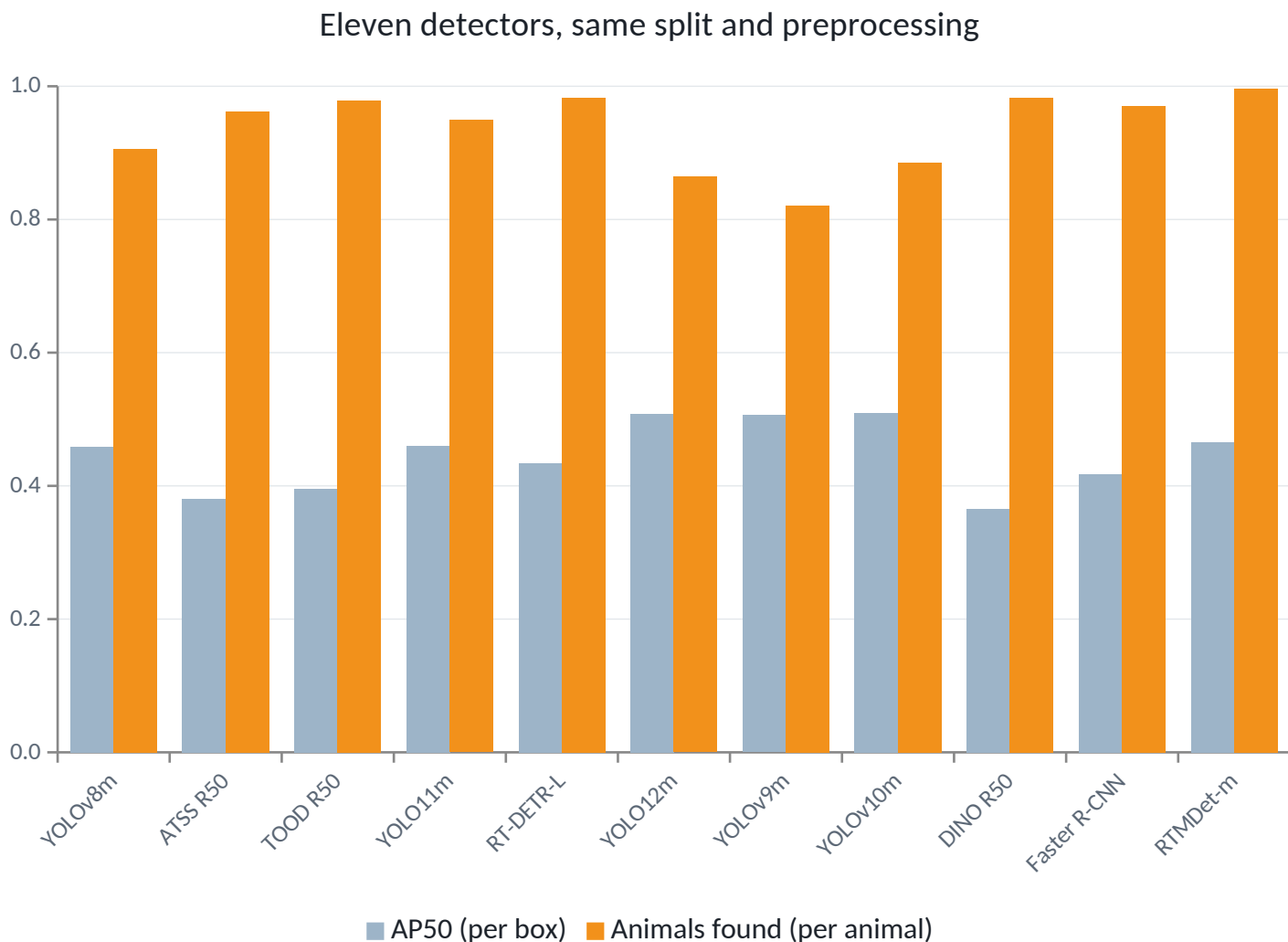
Did the confirmation rule accept a track on the animal?

Measures confirmation

Held-out protocol

The rule is fitted by grid search on the 19 training videos, frozen, and reported on 13 videos the detector never saw (83 deer).

Detection is close to solved; the count is lost afterwards



82 to 99.6%

of animals are found in at least one frame by every detector, although AP50 is only 0.37 to 0.51

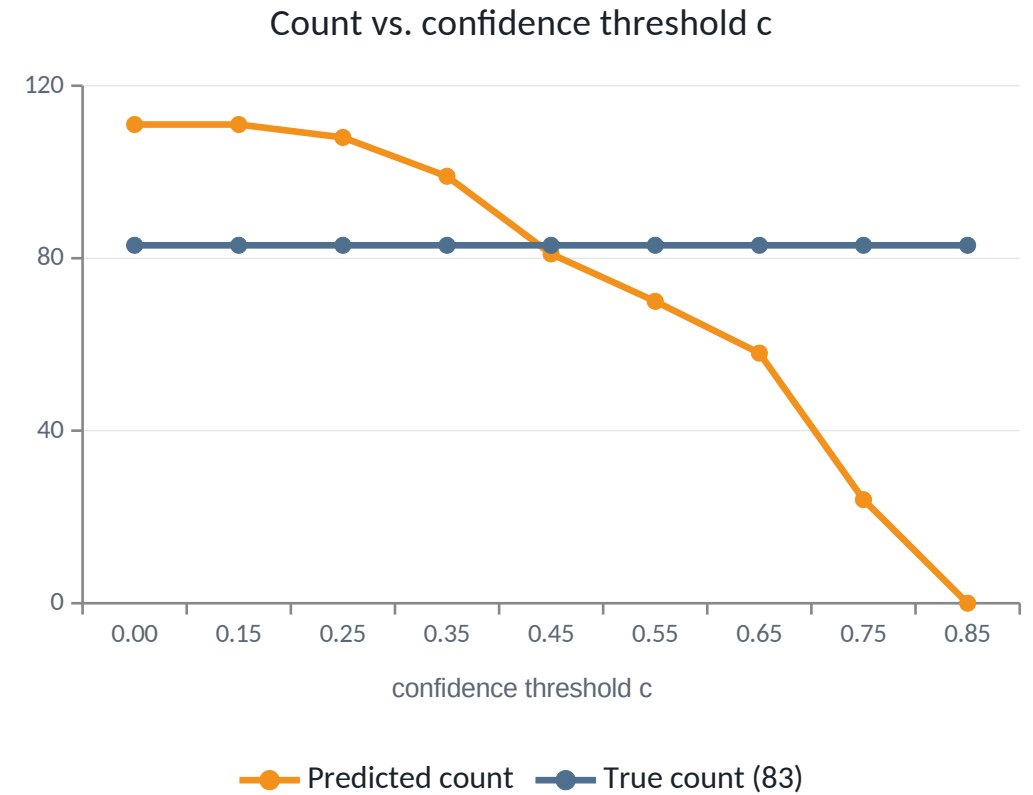
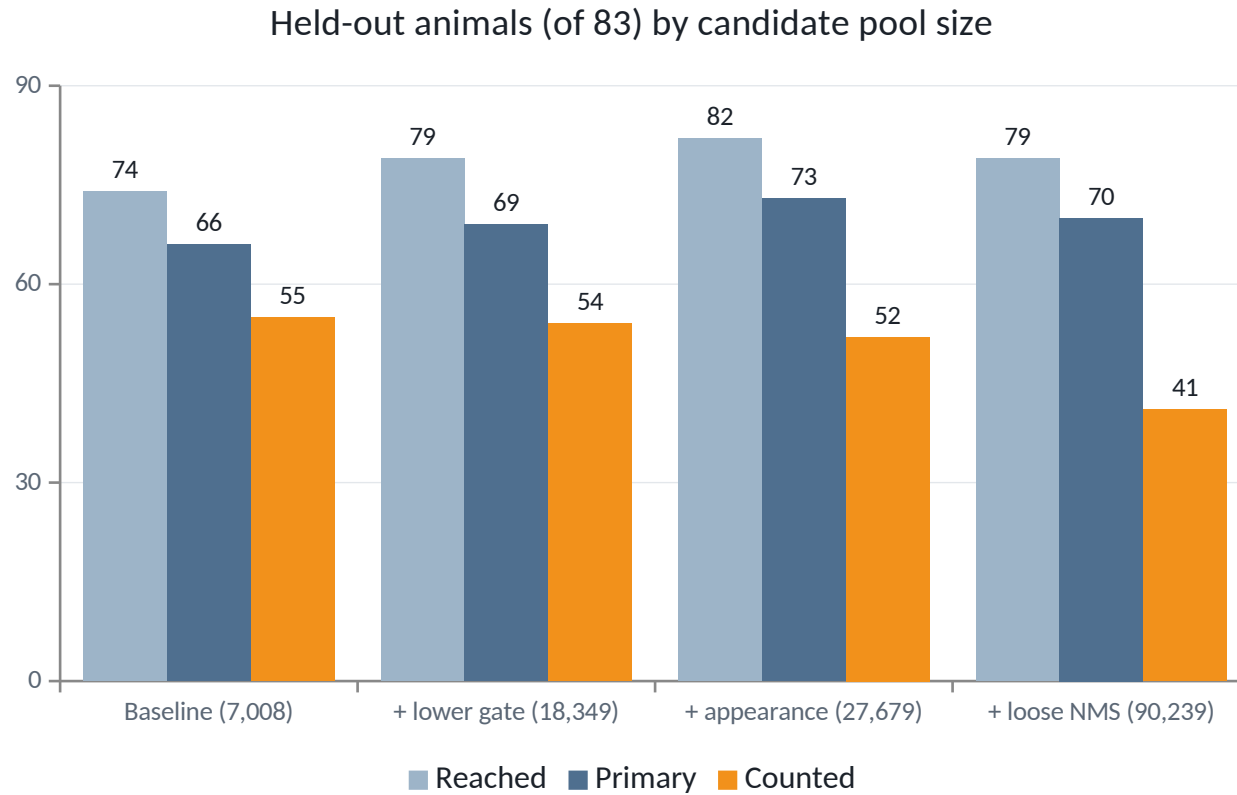
89 > 80 > 66%

reached, primary and counted on held-out video: most animals are lost at confirmation

MAE 2.38

animals per video; the pipeline under-counts and almost never over-counts

Richer candidate pools count fewer animals

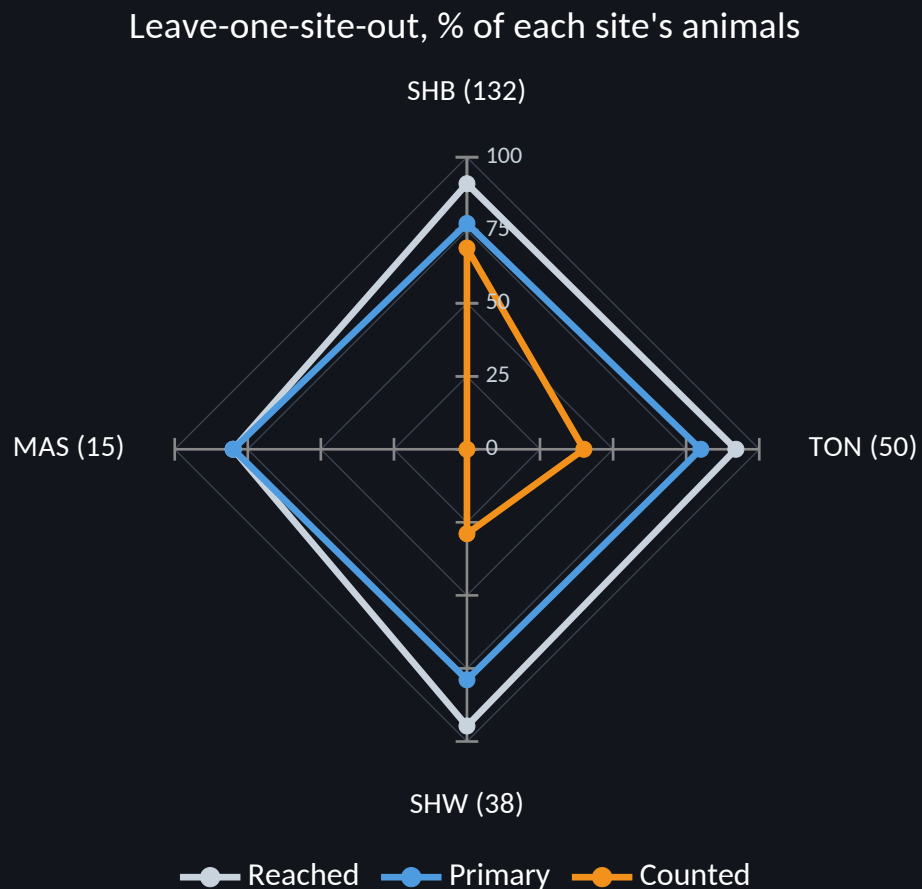


Every richer pool reaches more animals than the baseline and counts fewer: the extra candidates are re-sightings of animals already counted.

Lowering c recovers animals but admits about two duplicates for each; the count is unbiased near $c = 0.45$ (81 of 83).

Learned confirmers (logistic regression, boosting, transformer) lose to the 3-parameter rule: MAE 2.85 to 3.00 vs. 2.38.

Across sites, and what to take away



Detection and association transfer to an unseen site; the count does not. An absolute threshold of 0.65 accepts nothing at MAS, where no track scores above 0.585.

1

Detection is not the bottleneck

Per animal, detection is near-saturated on this corpus. Confirmation is where animals are lost.

2

Improving the candidate pool did not help

Better detection, association and looser suppression each left the count equal or lower.

3

The limit is supervision, not model capacity

About 200 positive tracks; duplicates sit between true animals and false candidates on every feature.

4

Choose the operating point for the purpose

Minimum error per transect ($c = 0.65$) or an unbiased total for abundance ($c = 0.45$); re-fit at a new site.

Dataset, annotations and evaluation code will be released with the paper.